
Development of PLC based Several Automatic Measurement and Monitoring System.

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BACHELOR OF SCIENCE IN ELECTRICAL & ELECTRONIC
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Declaration of Authorship

I, Mohammad Badal Ahmmed, declare that, this thesis titled, ‘**Development of PLC based Several Automatic Measurement and Monitoring System.**’ and the work presented in it at my own. I confirm that this work was done totally or mainly at this University. Any part of this thesis has previously not been submitted for a degree or any other qualification at this University or any other institution. I have clearly quoted the published work of others which have been consulted in this thesis. I also have acknowledged all main sources of help.

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Abstract

Programmable Logic Controller (PLC) is the typical and is most useful at industrial field. It is invented to replace the necessary sequence relay circuit for machine control. In machining, packaging, materials handling, automated assembly or countless other industries applications the PLC is most efficient considering money and time than any other control. As more complex functions are used the more complex model of the PLC must be used. The instruction is given by Programming with Ladder Logic. Fortunately most PLCs have software which converts Ladder Diagram into Code and is work by continuously scanning the program. In any production line, machine function or process can be greatly enhanced using this type of control system. However, the biggest benefit is using a PLC is the ability to change and replace the operation or process while collection and communicating vital information. The modification of programming is possible without changing the components. They have the advantages of an all in one package, typically with display and buttons. Supervisory Control And Data Acquisition (SCADA) system for example Human Machine Interface (HMI) makes is more preferable for real time control of the process parameters. A PLC these days can complete price wise and functionally with this controllers especially if more than one controller is needed. Unlike a personal computer through the PLC is designed to survive in a rugged industrial atmosphere and to be very flexible in how it interfaces with inputs and outputs to the real world. PLC is on the right task as an industrial controller and increases a wide range of use. Thus it is one of the most typical factory automating controllers.

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Abbreviations

PLC	P rogrammable L ogic C ontroller
SCADA	S upervisory C ontrol A nd D ata A cquisition
HMI	H uman M achine I nterface
LAD	L adder D iagram
SFC	S equence F unction C hart

Dedicated to my Friends...

Chapter 1

Introduction

1.1 Introduction

Industrial control system (ICS) is a general term that encompasses several types of control systems, including supervisory control and data acquisition (SCADA) systems, distributed control systems (DCS), and other control system configurations such as skid-mounted Programmable Logic Controllers (PLC) often found in the industrial sectors and critical infrastructures. ICS are typically used in industries such as electrical, water and waste water, oil and natural gas, chemical, transportation, pharmaceutical, pulp and paper, food and beverage, and discrete manufacturing (e.g., automotive, aerospace, and durable goods.) [1]

Equipment used in industrial control, such as a programmable logic controller (PLC), has increasing levels of connectivity and embedded processing. Additionally, there is a desire to make machinery functionally safe to operate. Delving deeper into the architecture and design of I/O modules in these systems, it is observed that a mutually reinforcing cycle between advances in automation and proliferation of electronic technology.[2]

A PLC is a microprocessor-based control system that uses a programmable memory for the internal storage of user orientated instructions for implementing specific functions such as arithmetic, counting, logic, sequencing, and timing.[3, 4] PLC can communicate in network with other PLC, remote I/O module, CAN communication with several PLCs and can constitute DCS with other intelligent devices. PLC proves themselves as a very versatile and effective tool in industrial control of electric drives. [5]

1.2 Motivation

As the number of people increasing, their needs also increasing. To fulfill their needs various industries will grow up day after day. So there need a sophisticated, smooth, easy handling controller for the industries. PLC provides all the required terms that an automated industries needed. The importance of PLC and programming with Ladder Diagram for the industrial automation are the main motivation of these thesis. My goal is to design an automated control system and develop a **Supervisory Control And Data Acquisition (SCADA)** system.

1.3 Objectives of the Thesis

In this thesis PLC and SCADA based automated control systems for Automated Industry were studied and developed that have a lot of application in the industries in our country. The main objectives of the thesis are given below:

- Familiarization with the PLC controlled Industrial Automation.
- Importance of PLC over any other controller.
- Study about Hardware and Programming Features of the Programmable Logic Controller.
- Study Supervisory Control and Data Acquisition (SCADA) System.
- Practical Implementations by PLC:
 1. Temperature Monitoring and Control.
 2. Mass/Seeking Control.
 3. Automatic Water Level Controller.
 4. Linear Distance and Velocity Measurement.

1.4 Organization

In this paper firstly the basics control system were described in **Chapter 1**. . Some important terms related with control system were listed with some examples which is necessary to understand the PLC based automatic industry and their application.

Chapter 2

Introduction to Control System Technology

2.1 Introduction

In these Chapter the basics of control system are described which are necessary to understand any kind of control system. There are some important terms like control concept, process parameters, different types of control system and methods are described . A method to choose the control system for any industry is also listed in the last of these chapter.

2.2 Control Concept

Control problems exist and arise in numerous fields of human activity. Controlling a process (an object) can be defined as influencing it in such a way as to force it to operate in accordance with certain assumed requirements. This definition applies to all processes which undergo control. A controlled process is always surrounded by the environment in which it exists, undergoing controlled or uncontrolled influences of this environment. The controlled influences are generated by a control unit, e.g., in a form of algorithms executed by an automatic control computer or in a form of decisions made by human beings.

The process input variables, whose values can be changed by the control unit are usually called the process manipulated variables or the process control inputs. Evaluation of the state of a controlled process is done on the basis of observations of values and features

of appropriate variables characterizing the process behavior. These variables are called process output variables. The general structure of a control system is presented in 2.1. There are two basic methods of decomposition of the overall control objective [6, 7]

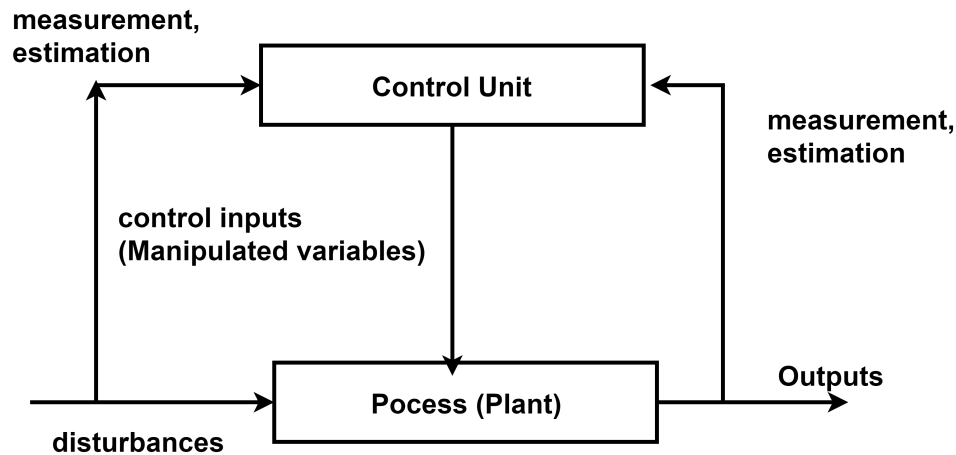


FIGURE 2.1: General control system structure

- Functional decomposition
- Spatial decomposition

The functional decomposition applies to a process treated as a whole and is based on assigning a set of functionally different partial control objectives in a structure of vertical, hierarchical dependence, called the multilayer control structure. A decision unit connected with each layer makes decisions concerning the controlled process, but each of them makes decisions of a different kind. On the other hand, the spatial decomposition is connected with a spatial structure of a complex (large-scale) controlled process. The multilayer control structure is a result of a functional decomposition of the general basic control objective. The realization of the basic, economic objective of the on-line control of an industrial (technological) plant can be expressed as the realization of a number of partial objectives. The three most important are:

1. Ensuring a safe running of the processes in the controlled plant, i.e., limiting the possibility of emergency situations to an acceptable level.
2. Ensuring required features of the process outputs (quality of products, etc.), i.e., maintaining the output variables within ranges of acceptable values.

3. Optimization of effectiveness of the process operation, usually maximization of the product value (under restrictions on usage of raw materials or utilities) or minimization of production costs at an assumed level of production, over a long time horizon.

2.3 Process Parameters

The physical variables which is measured and sometimes controlled are known as process parameters. An example of this would be the temperature of a furnace. The current temperature is called the process variable, while the desired temperature is known as the set-point. The process parameter that is controlled in manufacturing process are:

- Pressure
- Level
- Temperature
- pH
- Conductivity
- Humidity
- Density
- Linear distance, velocity
- Vibration
- Angular velocity , speed etc.

2.3.1 Necessity of Measurement

Measurement of process variables are important in controlling a process. The process variable is a dynamic feature of the process which may change rapidly. Accurate measurement of process variables is important for the maintenance of accuracy in a process. So, it is necessary to measure different variables for the better control of process and to get best product/output. There are four commonly measured variables which affect chemical and physical process: pressure, temperature, level and flow. [

2.3.2 Process Parameter Control

Keeping the process parameters or physical variables or process variables constant as per desired value is called process parameter control. This is generally done by measuring the instantaneous value of physical variables, comparing it with the desired value and using the difference to make a correction, which reduces this difference.

2.4 Different Types of Controls

There are different types of control systems exists in the industrial application. First the control system can be classified by two methods.

1. Manual Control
2. Automatic Control

2.4.1 Manual Control

Manual controls are performed by individuals outside of a system. In this control system the operator controls the process parameter of the plant. Everything depends upon the operator. This type of control has a lot of disadvantages because of low reliability, low process rate, unable to control complex system etc.

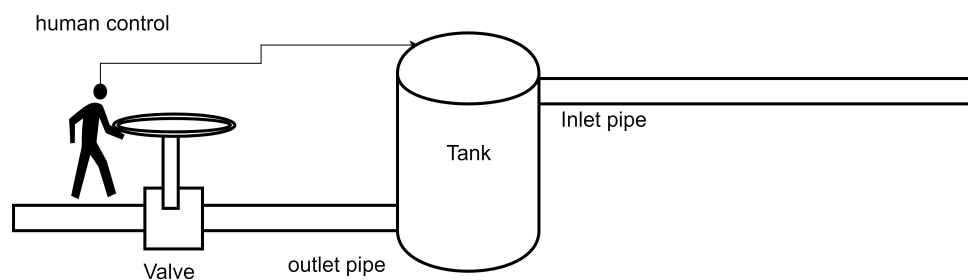


FIGURE 2.2: Manual control system example

2.4.2 Automatic Control

Automatic control is the application of control theory for regulation of processes without direct human intervention. In the simplest type of an automatic control loop, a controller

compares a measured value of a process with a desired set value, and processes the resulting error signal to change some input to the process.[8] The advantages of automatic

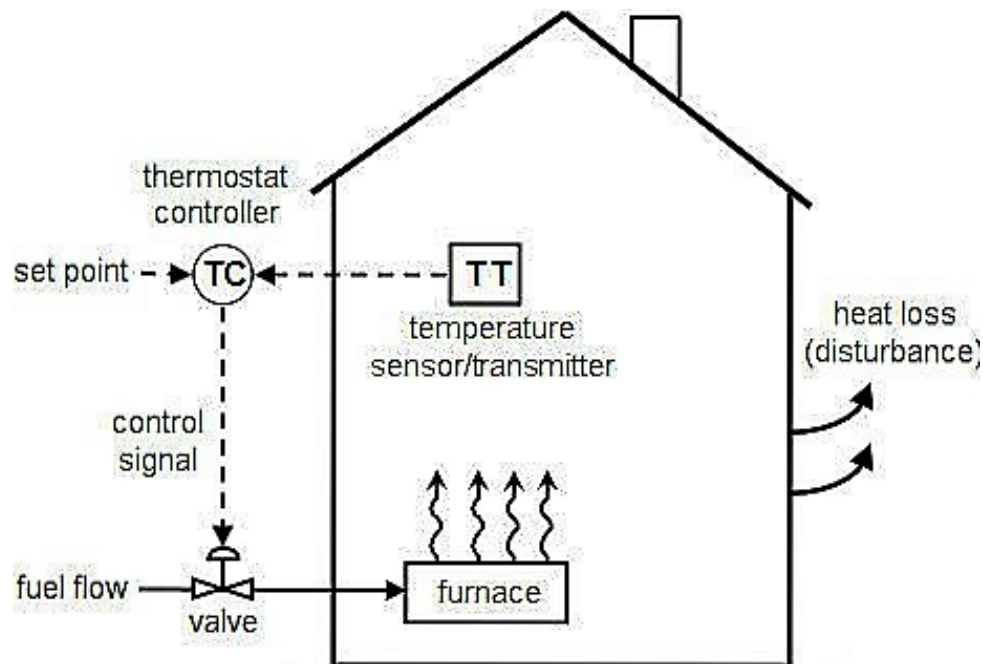


FIGURE 2.3: Automatic control system example[9]

Control are:

1. Increased throughput or productivity.
2. Increased consistency of output.
3. Improved robustness (consistency), of processes or product.
4. Improved quality or increased predictability of quality.
5. Reduced direct human labor costs and expenses.
6. Reduces operation time and work handling time significantly.

Depending on the loop control system can be classified by

1. Open loop Control
2. Closed loop Control

2.4.3 Open Loop Control

Open-loop system, referred to as non-feedback system, is a type of continuous control system in which the output has no influence or effect on the control action of the input signal. In other words, in an open-loop control system the output is neither measured nor feed back for comparison with the input. Therefore, an open-loop system is expected to faithfully follow its input command or set point regardless of the final result. Also, an open-loop system has no knowledge of the output condition so cannot self-correct any errors it could make when the preset value drifts, even if this results in large deviations from the preset value.

Another disadvantage of open-loop systems is that they are poorly equipped to handle disturbances or changes in the conditions which may reduce its ability to complete the desired task. For example, the dryer door opens and heat is lost. The timing controller continues regardless for the full 30 minutes but the clothes are not heated or dried at the end of the drying process. This is because there is no information fed back to maintain a constant temperature.

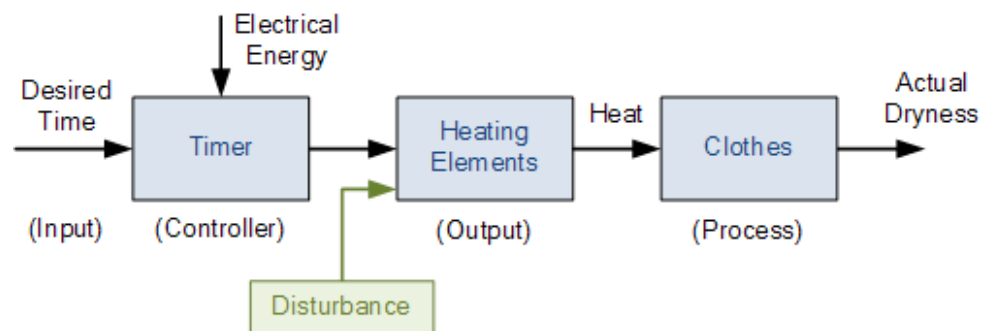


FIGURE 2.4: Open Loop control system example[10]

2.4.4 Closed Loop Control

A Closed-loop Control System, also known as a feedback control system is a control system which uses the concept of an open loop system as its forward path but has one or more feedback loops (hence its name) or paths between its output and its input. The reference to feedback, simply means that some portion of the output is returned back to the input to form part of the systems excitation.

Closed-loop systems are designed to automatically achieve and maintain the desired output condition by comparing it with the actual condition. It does this by generating an error signal which is the difference between the output and the reference input. In other

words, a closed-loop system is a fully automatic control system in which its control action being dependent on the output in some way.

So for example, consider our electric clothes dryer from the previous open-loop tutorial. Suppose we used a sensor or transducer (input device) to continually monitor the temperature or dryness of the clothes and feed a signal relating to the dryness back to the controller as shown in 2.5

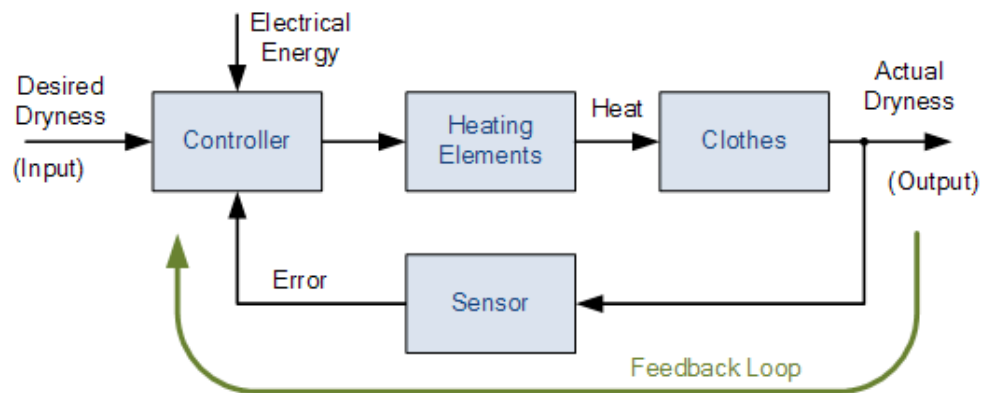


FIGURE 2.5: Closed Loop control system example[10]

2.5 Different Methods of Control Technology

There are different types of Control techniques are exists in industry. Like

1. Conventional Control System
2. Computerized Control System
3. Distributed Control System
4. PLC Based Control System

2.5.1 Conventional Control System

Conventional controls include simple electromechanical devices such as on-off thermostats, pressure switches, single and multi-stage fan controls, and timers for light scheduling. Self-contained, solid-state controls are also available for these types of tasks. Most conventional electromechanical controls have sensors which must be located close to the control, they do not have displays for the controlled variable, and the user who wants to

know how well the control is working must use a second device to monitor the conditions (e.g., a thermometer may be required for the typical electromechanical thermostat). In contrast, conventional solid-state controls do have displays for the controlled variable, but it is usually only a local display.

Advantages of conventional over computer-based controls include:

1. Conventional controls, especially the electromechanical ones, tend to be less susceptible to damage by electrical surges (although computer-based systems can be protected by appropriate surge suppressors).
2. Conventional controls are more readily available, easy to replace by unskilled labor and relatively cheap.

2.5.2 Computerized Control System

The computerized control system ensures quality production through real-time monitoring of all components of the mixture. The system is mainly characterized by the following:

1. Process automation
2. Virtual panel manual
3. Integrated handling
4. Virtual discharge sequence
5. Automatic moisture compensation
6. Intelligent humidity control
7. Intelligent recycling
8. User friendly interface
9. Open Process Control
10. Integrated plant simulator
11. Remote monitoring

Computer-based controls have several advantages over conventional controls including the following:

1. They can be programmed for almost any desired type of control including simple on-off, proportional, time proportioning, diurnal (e.g., a light program that varies lights in a house on a cycle similar to natural outdoor light), etc.
2. The control program can be set up with numerous logical interactions, e.g., turn the evaporative cooler on if the temperature is $\geq 80^{\circ}\text{F}$ and the humidity is $\leq 70\%$ or if the temperature is $\geq 85^{\circ}\text{F}$ regardless of the humidity.
3. Changes in control strategy require only a change in the program (software), whereas similar changes in conventional controls require re-wiring the system.
4. The computer-based control system can incorporate local and remote text, graphics, or animation displays of controlled or monitored conditions at selectable intervals. It can also record the conditions on removable or permanent disks, and on a print out.
5. The computer-based control system can be designed for phone access or networked to other computers by wired or wireless systems including one in the home.
6. The computer-based control system can be expanded to include other sensors, functions, houses, etc. in the future.
7. The computer-based control system has a high reliability and very few or not moving parts.

2.5.3 Distributed Control System

Distributed control systems (DCS) use decentralized elements or subsystems to control distributed processes or manufacturing systems. They offer flexibility, extended equipment life, simplicity of new equipment integration, and centralized maintenance when used in an industrial environment. A distributed control system involves the placement of multiple controllers within a plant or manufacturing process. The controllers are networked to a central console. DCSs aim to centralize plant operations to allow control, monitoring, and reporting of individual components and processes at a single location. DCSs are by definition hierarchical systems, although not all systems share an identical hierarchy.

The figure 2.6 below shows a typical DCS. Individual controllers, supervised by master controllers, make up the lowest "field" or "plant" level of the hierarchy. The master controllers connect to individual computers and servers, which are further connected to video output devices and a human machine interface (HMI), which is the actual point of

user control. DCSs are usually networked using standard protocols such as PROFIBUS and Ethernet, the latter of which is used in this particular system.

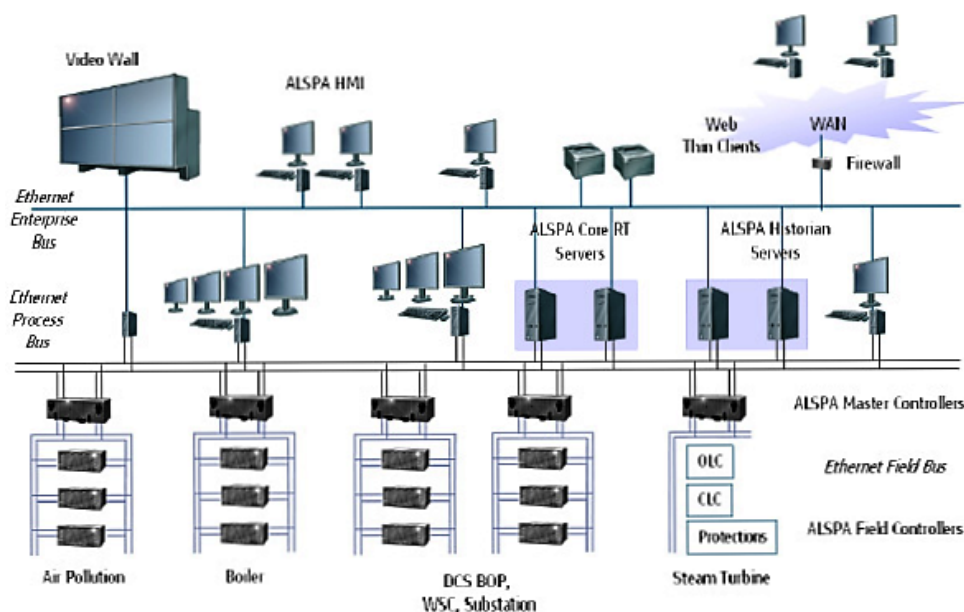


FIGURE 2.6: Distributed control system example[11]

2.5.4 PLC Based Control System

Programmable Logic Controllers (PLC) continue to evolve as new technologies are added to their capabilities. The PLC started out as a replacement for banks of relays. Gradually, various math and logic manipulation functions were added. Today they are the brains of the vast majority of automation, processes and special machines. PLCs now incorporate smaller cases, faster CPUs, networking and various internet technologies.

While PLCs were very good at quickly controlling automation, they did not share data easily. At best, PLCs would exchange information with operator interfaces (HMI) and Supervisory Control and Data Acquisition (SCADA) software packages on the factory floor. Any data exchange with the Business Level of the company (information services, scheduling, accounting and analysis systems) had to be collected, converted and relayed through a SCADA package. Typical of most PLCs, the communication networks were unique to the brand and limited in speed. With the acceptance of Ethernet, communication network speeds have increased but are still sometimes using proprietary protocols. Overall, PLCs are getting faster, smaller, cheaper and more powerful. As a result, they are gaining capabilities that used to be the exclusive domain of the Personal Computer (PC) and workstation arena. This translates into critical data quickly and cheaply being

shared directly between the PLCs on the Factory Floor and the Business Level of the company. Some of the features that a PLC can bring to your automation projects are Web Servers, FTP Servers, sending E-mail and Internal Relational Databases.

2.6 Selection of Control System

Considerations that need to be taken into account prior to selecting a control and monitoring system for environmental control include the following:

1. What control accuracy is needed?
2. Will the set points be fixed, cyclic or stepped?
3. How and where will the controlled variables (e.g., temperature, humidity, static pressure, light, etc.) be sensed by the controls and how will they be monitored and/or recorded?
4. Do the displays need to be local or remote or both?
5. Is it desirable to record some of the environmental parameters like temperature, humidity, and light level on a permanent record that can be used to develop trend graphs, etc.?
6. Is it likely that the producer will want to change the control strategy in the near future, and if so, will he/she be willing to re-wire the system to accomplish the new control strategy?
7. What is the potential for improving performance and minimizing losses with optimized control and monitoring of environment?
8. What is the potential for energy savings with optimized control?
9. Considering the risks of failure in critical parts of the system, what level of reliability is required?
10. What level of user friendliness is required?

Chapter 3

An Overview of Programmable Logic Controller

3.1 Introduction

Programmable Logic Controller is a revolution in control system engineering. It is used in industry as a logic controller to control the process more efficiently, reliably with low cost and low disturbance. In this chapter the history of Programmable Logic Controller, manufacturers of PLC, comparison with other types of conventional control system, application of PLC, advantages and disadvantages of PLC are described.

3.2 Programmable Logic Controller (PLC)

The programmable logic controller is the replacement of conventional sequence system such as electromagnetical relays, timers, counters etc. A PLC is a microprocessor based electronic environment designed for used in automatic control system. A Programmable Logic Controller (PLC) is an industrial computer control system that continuously monitors the state of input devices and makes decisions based upon a custom program to control the state of output devices.

PLCs and their associated peripherals are intended to be used in an industrial environment and may be provided as open or enclosed equipment. If a PLC or its associated peripherals are intended for use in other environments, then the specific requirements, standards and installation practices for those other environments must be additionally applied to the PLC and its associated peripherals.[12]

The functionality of a programmable controller can be performed as well on a specific hardware and software platform as on a general-purpose computer or a personal computer with industrial environment features. This standard applies to any products performing the function of PLCs and/or their associated peripherals. This standard does not deal with the functional safety or other aspects of the overall automated system. PLCs, their application programme and their associated peripherals are considered as components of a control system.

Almost any production line, machine function, or process can be greatly enhanced using this type of control system. However, the biggest benefit in using a PLC is the ability to change and replicate the operation or process while collecting and communicating vital information.

The sheer number of PLC applications is enormous. According to a recent Control Engineering magazine poll, The major applications for PLCs include machine control (87%), process control (58%), motion control (40%), batch control (26%), diagnostic (18%), and other (3%). The results don't add up to 100% because a single control system generally has multiple applications.[13]

3.3 History of PLC

The early history of the PLC goes back to the 1960s when control systems were still handled using relay control. During this time the control rooms consisted of several walls containing many relays, terminal blocks and mass of wires. The problems related to such kind of systems were many, among those one could mention:

1. The lack of flexibility to expand the process, as well as the inordinate amount of time needed to adjust the process when changes were needed.
2. Troubleshooting which covered from dirty contacts, loose wires, outdated prints on the terminal blocks with informal nomenclature, to cryptic connection documentation.

These problems were faced continuously by technician and control engineers.

In 1968 Bill Stone, who was part of a group of engineers at the Hydramatic Division of General Motors Corporation, presented a paper at the Westinghouse Conference outlining their problems with reliability and documentation for the machines at this plant. He also presented a design criteria developed by the GM engineers for a standard machine controller. By the time of the proposal, people at Bedford Associates, which included

Richard Morley, Mike Greenberg, Jonas Landau, George Schwenk and Tom Boissevain, were already working on the design of a unit, which characteristics included a modular and rugged design, the use of no interrupts for processing, as well as direct mapping into memory.

The Bedford team named this unit as the 084, since it was the 84th project for the company. After finding some financial support, the team decided to form a new company called Modicon (MODular DIGital CONTroller) which worked closely with Bedford to create the controller. The team at Modicon was finishing the design and build of the 084, that now they were calling the programmable controller (PC).

Finally in 1969, the winning proposal came from Bedford Associates and Modicon, when they demonstrated at GM the Modicon 084 solid-state sequential logic solver. The Mod-



FIGURE 3.1: Manual relay panels from the early 1960s.

icon 084 consisted of three distinct components that included the processor board, the memory, and the logic solver board, which solved the dominant algorithms associated with ladder logic. When the Modicon 084 was designed, it was built rugged with no ON/OFF switch, totally enclosed with conductive cooling, it was designed to be reliable.[13]

By 1971, Odo Struger and Ernst Dummermuth engineers of Allen-Bradley, had begun to develop a new concept that included improvements based on the customer needs that could not be fulfilled by its second attempt, the PMC. This new concept was known as the Bulletin 1774 PLC. Allen-Bradley named this new device as the Programmable Logic Controller (PLC) over the then accepted term Programmable Controller. The PLC terminology became the industrial standard especially when PC became associated with personal computers. In 1985 Rockwell Automation acquired Allen-Bradley. The name

of the new produced products are still associated to Allen-Bradley. The early 1980s saw

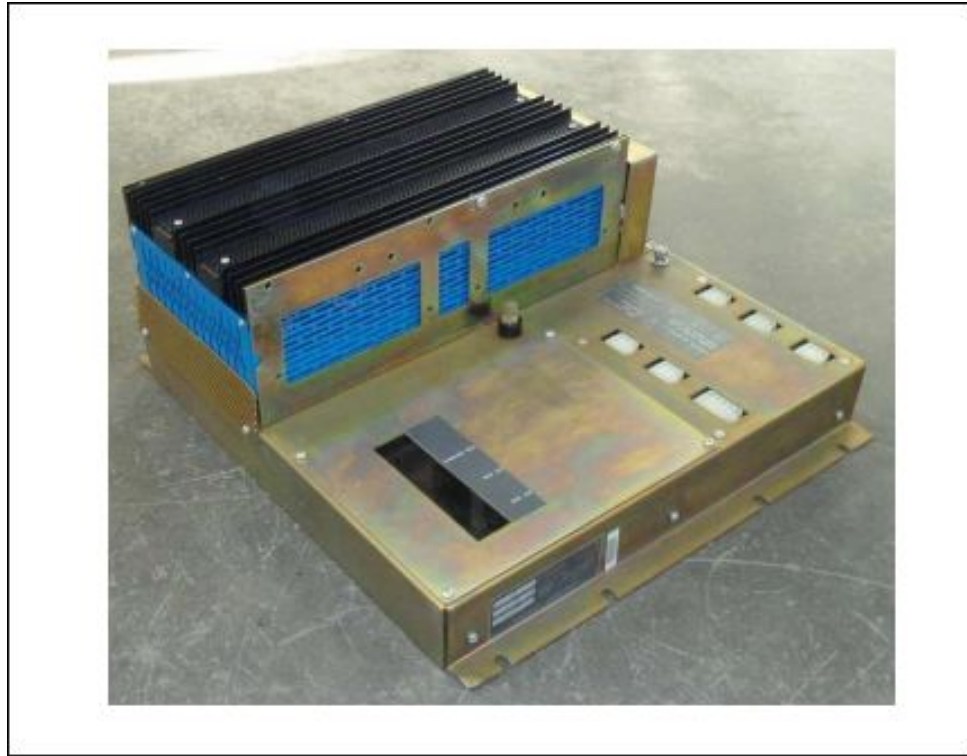


FIGURE 3.2: The first programmable logic controller of Allen Bradley, the Bulletin 1774 PLC. It was invented by Ernst Dummermuth in 1974.

a cross pollination between PLCs and distributed control systems (DCSs). Where PLCs already begun incorporating distributed control functions so they could be linked much in the way that DCSs were linked. Building on the trend, software companies sprang up in great numbers during this time. During the 90s, standardization and open systems were the main themes. Ethernet peer-to-peer networking became available from virtually all PLC manufacturers. EEPROM and Flash memories replaced the EPROMs of the 1980s. PCs and CRTs in general became accepted and started to replace switches and lights on control system panels. Small PLCs called Bricks were introduced to the marketplace. Redundancy for PLCs became a standard product. The first few years of the 21st century have seen a consolidation of PLC manufacturers. Very small nano or pico PLCs, some as small as industrial relays, have appeared. Safety PLCs featuring triple redundancy were introduced. LCD base operator interface panels have largely displaced CRTs, especially on the plant floor.

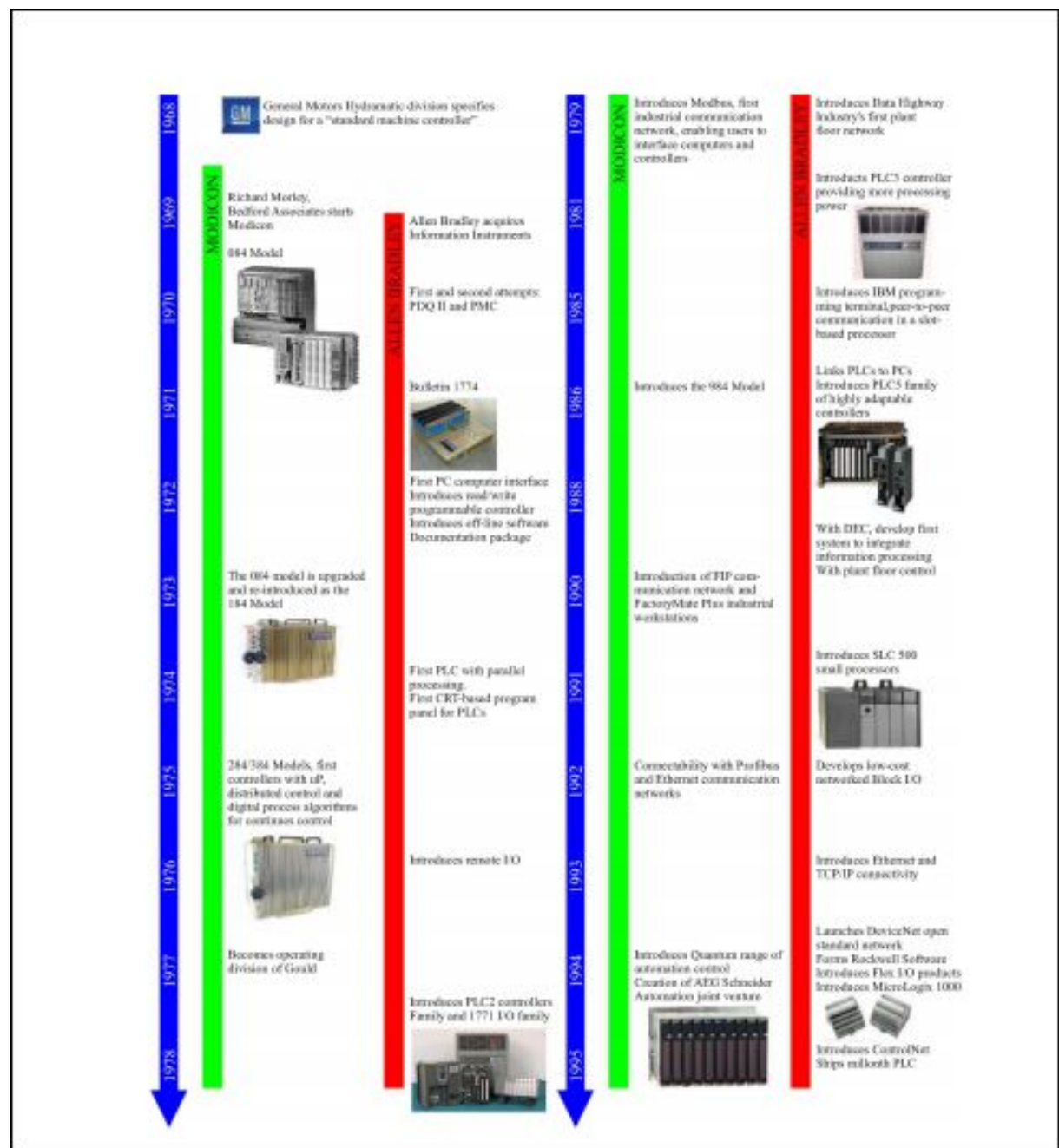


FIGURE 3.3: Modicon and Allen-Bradley PLC timeline (1968-1995)[13]

3.4 Manufacturers of PLC

Major PLC manufacturers are as follows:

1. Siemens
2. ABB
3. Schneider (Modicon)

4. Rockwell (Allen-Bradley)
5. Mitsubishi
6. GE
7. Omron
8. Bosch Rexroth
9. Beckhoff
10. Fuji
11. Toshiba

3.5 PLC vs. other types of Controls

TABLE 3.1: Conventional Control system Vs. PLC Based Control System

Parameter	Conventional Control system	PLC Based Control System
1.Wiring System	Hard Wiring	Software Programming
2.Space Requirement	Large Space and increased number of panel board Required	Smaller Space Required
3.Power Consumption	High	Low
4.Maintenance	Extensive	Little
5.Installation	Difficult	Easy
6.Modification Within the Control System	Difficult	Easy as the control system is software based
7.Reliability	Very much vulnerable to mechanical faults, wiring faults and other failures.	Much less chance of mechanical failures.
8.Diagnosis of the Problem	Very Difficult	Easy.

3.6 Application of PLC

A programmable logic controller (PLC) is an electronic device used in many industries to monitor and control building systems and production processes. Unlike PCs and

smartphones, which are designed to perform any number of roles, a PLC is designed to perform a single set of tasks, except under real-time constraints and with superior reliability and performance.

To meet the demands of harsh industrial environments, PLCs are designed to be extremely robust, often capable of withstanding extreme temperatures, humidity, vibration, and electrical noise. Logic controllers are commonly tasked with monitoring and controlling a very large number of sensors and actuators, and are therefore distinct from other computer systems in their extensive input/output (I/O) arrangements.

The major application of programmable logic controller are given below:

3.6.1 Applications of PLC in Cement Industry

Along with the best-quality raw materials, the accurate data regarding process variables, especially during mixing processes within the kiln, ensures that the output provided should be of the best possible quality. Nowadays a DCS with bus technology is used in the production and management industry. By using this existing DCS control system, the PLC is in user mode of SCADA. This mode comprises PLC and configuration software. This SCADA mode comprises the PLC and host computer. The host computer consists of slave and master station. The PLC is used for controlling the ball milling, shaft kiln and Kiln of coal.

3.6.2 Application of PLC in Glass Industry

From the year 1980 the Programmable-logic controllers are in use in the glass industry, and they are assembled bit by bit. PLCs are used mainly in every procedure and workshop for controlling the material ratio, processing of flat glasses, etc.

With the development of PLC and increasing demand in the real world, the control mode of the programmable-logic controller with an intelligent device is applied in the glass industry. In making of a float glass, PLC itself cannot finish some controlling tasks because of the complexity of the control system and processing of huge data. For the production of glass, we make use of bus technology to construct the control mode of a PLC with a distributed-control system. This control system deals with analog controlling and data recording; the PLC is also used for digital quality control and position control. This type of control mode is a big advantage for PLC and DCS for improving reliability and flexibility of the control system.

3.6.3 Wireless Control in Industrial Automation

In industrial monitoring and control process, the industrial process parameters like voltage, current and temperatures are monitored remotely by using wireless communication, and thus enable to control various final control devices like relays, motors, etc., with the use of a micro controller.

In the transmitter section, the wireless module is configured in such a way that it receives the data collected from the micro controller and sends it to the remote receiver. In this system, the micro controller is programmed to collect the data from an analog to digital converter that continuously monitors temperature, voltage and current parameters.

At the receiver side, the wireless module receives all the sent data from a wireless transmitter within the range of communication. This data is further transferred to the micro controller using an embedded circuitry wherein the micro controller program compares all these data parameters with predefined set limits. If any parameter exceeds its limit, then the micro controller sends command signals to a relay driver IC, which is responsible to operate different loads such as motors, relays, circuit breakers, etc. All these parameters information is also displayed on LCD display as a Human machine interface.

In this way, industrial parameters can be easily monitored and controlled through the short range low cost and low powered wireless communication technology.[14]

3.6.4 Application of PLC in Hospital Emergency Calling Device

As the rapid development of information technology and automation technology, which increases efficiency of industry greatly, is making the patients of hospital wards and the health care workers in the automatic association. Each bed of the hospital wards and nurse room need long-distance automatic call from time to time, which is easy for patients in urgent of medical signal, and the signal staff can also deal with relevant affairs quickly and efficiently. Before hospital emergency call devices use a single chip or integrated circuits to control, however, these systems have many disadvantages: high reliability, easy extension and weak communication network. Programmable logic controllers have been widely used in various trades and industries because it is reliable, secure, flexible, convenient and economical. At present, with the falling prices and widespread of it, the application of PLC is a very good technical solution to control the hospital call.[15]

3.6.5 Power plants automation and control

Power plants control and automation is always a challenge for industry because electric power has such an important roll in the development of civilization. Upgrading from first automation systems (electromagnetic relays - cable logic) to modern systems that use Programmable Logic Controllers (PLC) is an important stage in the development of automation systems used in power engineering (generation, transport, distribution, consumption), followed by cost reduction.

3.7 Advantages of PLC

The major advantages of Programmable Logic Controllers are given below:

1. Very fast
2. Easy to change logic i.e. flexibility
3. Reliable due to absence of moving parts
4. Low power consumption
5. Easy maintenance due to modular assembly
6. Facilities in fault finding and diagnostic
7. Capable of handling of very complicated logic operations
8. Good documentation facilities
9. Easy to couple with the process computers
10. Analog signal handling and close loop control programming
11. Counter, timer and comparator can be programmed
12. Ease operator interface due to colourographic and advisory system introduction

3.8 Disadvantages of PLC

The major disadvantages of Programmable Logic Controllers are given below:

1. There's too much work required in connecting wires.
2. There's difficulty with changes or replacements.
3. It's always difficult to find errors; And require skillful work force.
4. When a problem occurs, hold-up time is indefinite, usually long.
5. It is needed to learn a new language and syntax for interfacing with it, regardless of proprietary code.
6. Ladder logic is an analog of relay logic, but statement list is a far cry from relay logic, symbolically. The learning curve on PLC's is longer than relay logic.

Chapter 4

Functional Description of PLC

4.1 Introduction

A programmable logic controller is a special form of microprocessor based control system that uses a programmable memory to store instructions and to implement functions such as logic, sequencing, timing, counting, calculating etc. and processing of analog and digital signals in order to control machines and process. It is replacement of conventional control system such as electromagnetic relays, timers, counters etc.

The designers of PLC have programmed it for the user, so that the control program can be enter easily using a simple form of language like Ladder diagram, Instruction code etc. Input devices such as switches, push button, pressure switches etc. and output devices such as motors, lamps, solenoids, and relays etc. which are physically connected to the input and output interfaces of the PLC. Most of the PLC operates internally between 5 to 15 volts, but input and output process signal is higher, typically 24 to 240 volts.

The information within the plc is carried by means of digital signals which is compatible with the processor is 5 volts dc. PLC can handle one program in memory at a time. But the microprocessor can process a number of inputs and produce outputs simultaneously.

In this chapter the functional description of PLC, has been described. In the beginning of the hardware description of PLC has been elaborately described. Memory of the PLC , its way of operation and data communication have been denominated in middle of the chapter. And last of the chapter some notes to select the PLC for any plants have been listed.

4.2 Hardware Description of PLC

The basic architecture of a PLC consists of the main components-the processor module, the power supply, and the I/O modules. The processor module consists of the central processing unit (CPU) and memory. In addition to a microprocessor, the CPU also contains at least an interface to a programming device and may contain interfaces to remote I/O and other communication networks. The power supply is usually a separate module, and the I/O modules are separate from the processor. The types of I/O modules include discrete (on/off), analog (continuous variable), and special modules like motion control or high-speed counters. The field devices are connected to the I/O modules.

A typical PLC consists of various functional units(modules) as shown in figure4.9

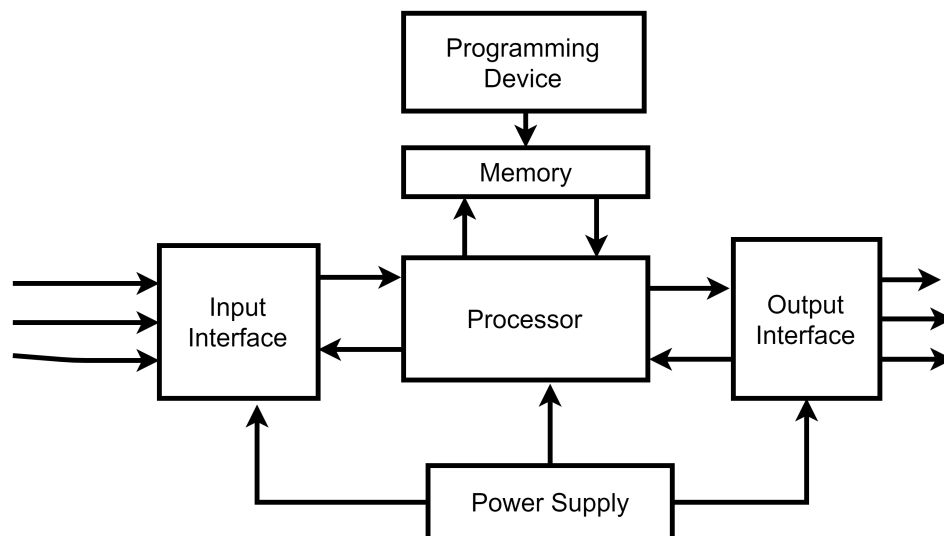


FIGURE 4.1: Functional block diagram of programmable logic controller

4.2.1 Power Supply(PSU)

The power supply unit is needed to convert the main AC voltage to the DC voltage (5V) necessary for the processor (CPU) and the circuits in the input and output interface modules.

4.2.2 Central Processing Unit (CPU)

The CPU is a microprocessor based system that replaces relays, timers, counters etc. It interprets the input signals and carried out the control action, according to the program stored by the user in its memory, communication the decisions as action signals to the

outputs. The CPU governs all the PLC activities.

The **memory module** is a major part of CPU. The memory unit is where the program is stored that to be used for the control actions to be executed by the microprocessor and data stored from the input for processing and for output for outputting. Memory Modules of PLC are as follows-

- RAM: Random Access Memory for program
- RAM: Random Access Memory for data
- ROM: Read only Memory
- PROM: Programmable Read Only Memory
- Memory Sub-Module: EPROM, EEPROM

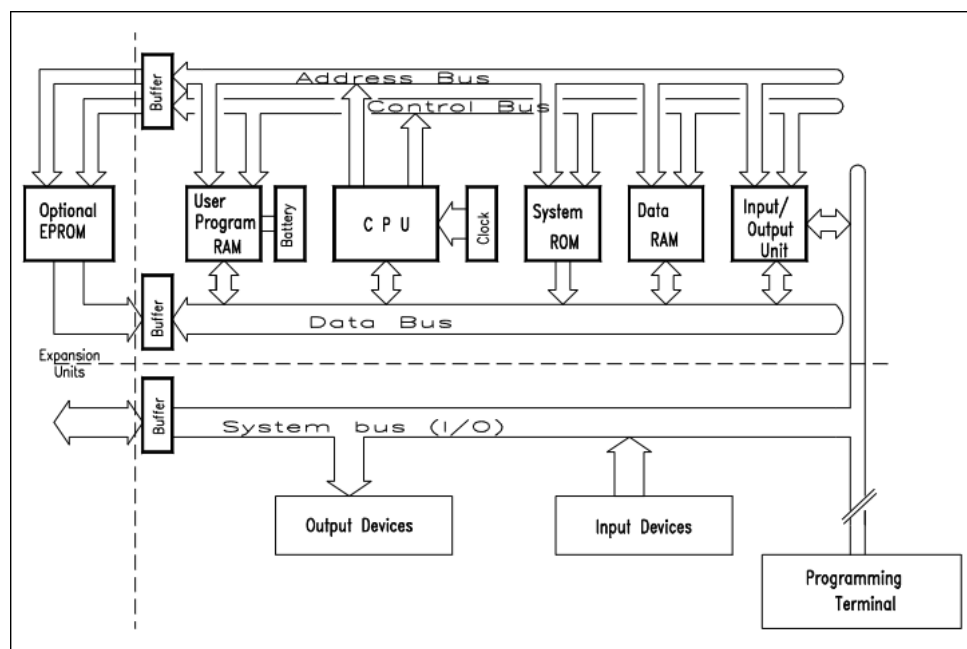


FIGURE 4.2: Functional block diagram of CPU of PLC

4.2.3 Input/Output Modules

PLC systems comprise input modules, output modules, and input/output modules. Since many of the inputs and outputs involve real-world analog variables while the controller is digital, PLC system hardware-design tasks focus on the requirements for digital-to-analog converters (DACs) and analog-to-digital converters (ADCs), input- and output signal-conditioning, and isolation of the electrical wiring of the input- and output modules from

the controller and each other.

Resolutions of I/O modules typically range from 12 bits to 16 bits, with 0.1% accuracy over the industrial temperature range. Analog output voltage and current ranges include 5 V, 10 V, or 0 V to 5 V, 0 V to 10 V, and 4 to 20 mA or 0 to 20 mA. Settling-time requirements for DACs vary from 10 s to 100 ms, depending on the application. Analog input ranges can be as small as 10 mV, from bridge transducers, as large as 10 V, from actuator controllers, or 4-to-20 mA currents, in industrial process-control systems. Conversion times, depending on the required accuracy and choice of ADC architecture, vary from 10 samples per second to hundreds of kilo-samples per second.

Digital isolators, optocouplers or electromagnetic isolators are used to isolate the ADCs, DACs, and signal-conditioning circuitry on the field side of the system from the controller on the digital side. If the system must also be fully isolated on the analog side, a converter would be necessary on each channel of the input or output to maximize the isolation between channels and isolated power would be necessary, via transformers or Analog Devices isoPower technology.

PLC-system analog outputs commonly used to control actuators, valves, and motors

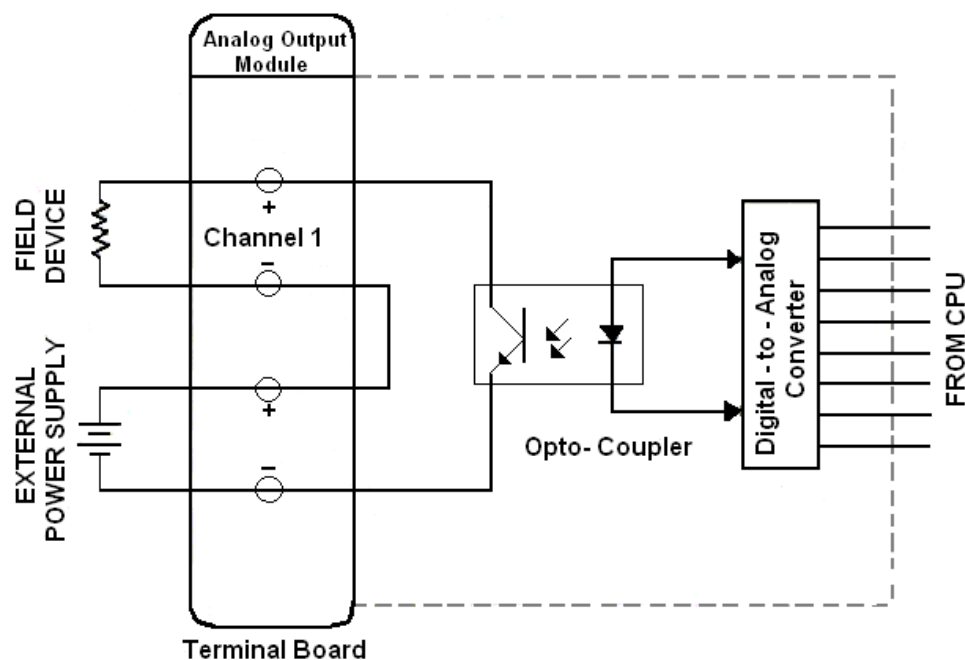


FIGURE 4.3: Functional block diagram of Input Module

in industrial environments employ standard analog output ranges such as 5 V, 10 V, 0 V to 5 V, 0 V to 10 V, 4 to 20 mA, or 0 to 20 mA. The analog output signal-chain often includes digital isolation to isolate the controller's digital outputs from the DAC and analog signal conditioning. Converters in digitally isolated systems predominantly use 3-wire or 4-wire serial interfaces to minimize the required number of digital isolators or optocouplers.

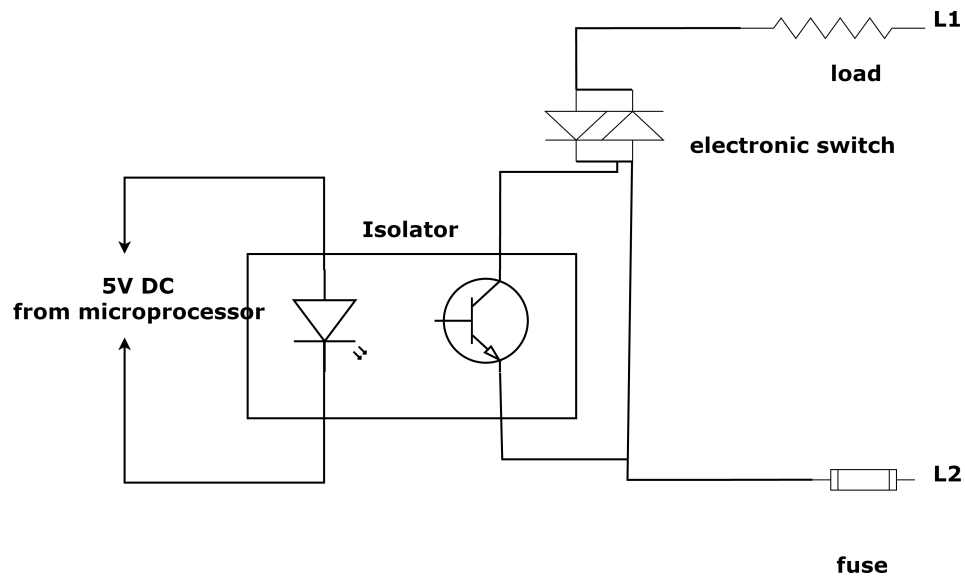


FIGURE 4.4: Functional block diagram of Output Module

4.2.4 Timers

Programmable Logic Controller has three types of timer. Such as-

1. On-Delay timer
2. Off-Delay timer
3. Retentive or Accumulating timer

On-Delay timer

This type of timer simply "delays turning on". In other words, after our sensor (input) turns on we wait x-seconds before activating a solenoid valve (output). This is the most common timer. It is often called TON (timer on-delay), TIM (timer) or TMR (timer).

Off-Delay timer

This type of timer is the opposite of the on-delay timer listed above. This timer simply "delays turning off". After our sensor (input) sees a target we turn on a solenoid (output). When the sensor no longer sees the target we hold the solenoid on for x-seconds before turning it off. It is called a TOF (timer off-delay) and is less common than the on-delay type listed above. (i.e. few manufacturers include this type of timer)

Retentive or Accumulating timer

This type of timer needs 2 inputs. One input starts the timing event (i.e. the clock starts ticking) and the other resets it. The on/off delay timers above would be reset if the input sensor wasn't on/off for the complete timer duration. This timer however holds or retains the current elapsed time when the sensor turns off in mid-stream. It is often called an RTO (retentive timer) or TMRA (accumulating timer).

4.2.5 Counters

A counter is set to some preset value and, when this value of input pulses has been received, it will operate its contacts. The counter accumulated value ONLY changes at the off to on transition of the pulse input. Typically counters can count from 0 to 9999, -32,768 to +32,767 or 0 to 65535. The normal counters are typically software counters they don't physically exist in the PLC but rather they are simulated in software. A good rule of thumb is simply to always use the normal (software) counters unless the pulses you are counting will arrive faster than 2X the scan time.

Counter Types

1. Up-counters counts from zero up to the preset value. These are called CTU, CNT, C, or CTR.
2. Down-counters count down from the preset value to zero. These are called CTD.
3. Up-down counters count up and/or down. These are called CTUD.
4. For CTU or CTD counter we need 2 inputs, but in CTUD we need 3 (up, down and preset).
5. High Speed Counter: Most manufacturers also include a limited number of high-speed counters (HSC). Typically a high-speed counter is a hardware device. Hardware counters are not dependent on scan time.

4.3 Input Output Device of PLC

4.4 Programming Device

The program is created in a programming device and then transferred to the PLC. The program can be created using a dedicated programming device if appropriate software is installed. A personal computer with appropriate software can also be used as a programming device. This device allows the user to enter, edit and monitor the programs by connecting it with the PLC CPU unit. Programming Device is usually not permanently connected to a particular PLC and can be removed from one PLC to the next without disturbing the operations.

Programming Devices are of two types:



FIGURE 4.5: Hand Programmer

1. Pocket/ Hand-Held Programmer
2. Personal Computer with appropriate software.

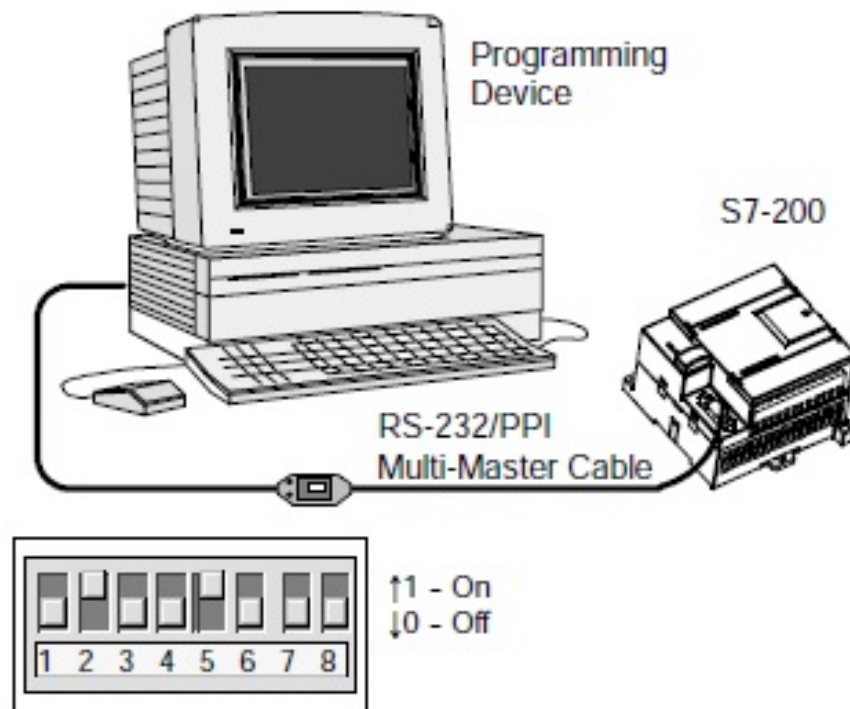


FIGURE 4.6: Personal Computer Programmer

The programming device can be used to select the various processor modes of operation. Some of common operation modes of PLC are:

1. Program Mode: It is used to enter a new program or update an existing one in the internal RAM memory. In this mode PLC does not execute the program.
2. Monitor Mode: This Mode allows a running program to be edited and monitored while connected during development.
3. Run Mode: It is used to execute the user program. Input unit are monitored and output devices are energized accordingly.

4.5 Mechanical Design of PLC system

There are two common types of Mechanical Design of PLC system. This are:

1. Single Box/Integral Type
2. Modular/ Rack type

4.5.1 Single Box/Integral Type

The box type is smaller and used for simpler control situations. It is supplied as an integral compact package, complete with power supply, processor, memory and input output units. Some of the most basic of these only have 4 outputs. They typically can have from 4 to 40 inputs and outputs. Box type PLCs have limited expansion capabilities.



FIGURE 4.7: Single Box/Integral Type PLC

4.5.2 Modular/ Rack type

The modular type consists of a central rack that houses various hand-picked modules that are appropriate for each control situation. A large variety of modules exist that satisfy many needs such as power supplies, processors, analog input and outputs, digital input and outputs, and communications. I/O modules can always be added after the unit is installed to suit new needs. Power supplies can be upgraded to meet new power requirements. Also the I/O modules can be much more specialized than that of the box type PLCs. For example, there are modules that serve as PID control or TTL logic inputs. Modular PLCs are used for larger and more complex operations. Typically they have from 20 to 100 inputs and outputs and up.



FIGURE 4.8: Modular/ Rack type

4.6 Data Communication

PLC I/O number can be increased through certain additional modules by system extension through extension lines. Each module can contain extension both of input and output lines. Extension modules can have inputs and outputs of a different nature from those on the PLC controller. When there are many I/O located considerable distances away from the PLC an economic solution is to use I/O modules and use cables to connect these, over the long distances, to the PLC. In some situations a number of PLCs may be linked together with a master PLC unit sending and receiving I/O data from the other units.

Twisted-pair cabling, often routed through steel conduit. Coaxial cable enables higher data rates to be transmitted and does not require the shielding of steel conduit. Fiber-optic cabling has the advantage of resistance to noise, small size and flexibility.

4.6.1 Parallel communication

Parallel communication is when all the constituent bits of a word are simultaneously transmitted along parallel cables. This allows data to be transmitted over short distances at high speeds. Might be used when connecting laboratory instruments to the system.

Parallel standards: The standard interface most commonly used for parallel communication is IEEE-488, and now termed as General Purpose Instrument Bus (GPIB). Parallel

data communications can take place between listeners , talkers , and controllers. There are 24 lines: 8 data (bidirectional), 5 status & control, 3 handshaking, and 8 ground lines.

4.6.2 Serial communication

Serial communication is when data is transmitted one bit at a time. A data word has to be separated into its constituent bits for transmission and then reassembled into the word when received. Serial communication is used for transmitting data over long distances. Might be used for the connection between a computer and a PLC.

Serial standards: RS-232 communications is the most popular method of PLC to external device communications. RS 232 is a communication interface included under SCADA applications. Other standards such as RS 422 and RS 423 are similar to RS-232 although they permit higher transmission rates and longer cable distances.

There are 2 types of RS-232 devices:

1. DTE Data Terminal Equipment and a common example is a computer.
2. DCE Data Communications Equipment and a common example is a modem.

4.7 PLC network

Modern control and business systems require open, digital communications. Industrial networks replace conventional point-to-point RS-232, RS-485, and 4-20 mA wiring between existing measurement devices and automation systems with an all-digital, 2-way communication network. Industrial networking technology offers several major improvements over existing systems. With industry-standard networks, we can select the right instrument and system for the job regardless of the control system manufacturer. Other benefits include:

1. Reduced wiring – resulting in lower overall installation and maintenance costs
2. Intelligent devices – leading to higher performance and increased functionality such as advanced diagnostics
3. Distributed control – with intelligent devices providing the flexibility to apply control either centrally or distributed for improved performance and reliability

4. Simplified wiring of a new installation, resulting in fewer, simpler drawings and overall reduced control system engineering costs
5. Lower installation costs for wiring, marshaling, and junction boxes

Standard industrial networks offer the capability to meet the expanding needs of manufacturing operations of all sizes. As our measurement and automation system needs grow, industrial networks provide an industry-standard, open infrastructure to add new capabilities to meet increasing manufacturing and production needs. For relatively low initial investments, we can install small computer-based measurement and automation systems that are compatible with large-scale and long-term plant control and business systems.[16]

4.7.1 Profibus

Profibus is commonly found in Process Control and large assembly, and material handling machines. Single-cable wiring of multi-input sensor blocks, pneumatic valves, complex intelligent devices, smaller sub-networks (such as AS-I), and operator interfaces.

Advantages: Profibus is the most widely accepted international networking standard. Nearly universal in Europe and also very popular in North America, South America, and parts of Africa and Asia. Profibus can handle large amounts of data at high speed and serve the needs of large installations. The DP, FMS and PA versions collectively address the majority of automation applications.

Disadvantages: High overhead to message ratio for small amounts of data; no power on the bus; slightly higher cost than some other buses; European and Siemens centricity is occasionally an obstacle for some North American users.

Profibus substantial speed, distance and data handling capabilities make it ideal for many process control and data intensive applications. Profibus DP, which is the most commonly messaging format for I/O, is a polling network, meaning that its assigned master periodically requests the status of each node. This ensures that each device on the network (which can send up to 244 bytes of data per scan) is updated consistently and reliably. Each message contains 12 bytes of overhead for a maximum message length of 256 bytes.

4.7.2 Controller Area Network (CAN)

CAN is the foundation for several popular field-level networks: DeviceNet, CANopen, SDS and others. When the developers of DeviceNet, CANopen, SDS and other CAN based networks sought a bulletproof technology for critical factory networking purposes, they opted for a ready made solution from the automotive industry. Bosch developed CAN in the early 1980 for eliminating large and expensive wiring harnesses in Mercedes automobiles. CAN was developed so that the primary control components in an automobile brake lights, airbags, sensors, lights, electric windows and door locks, etc. could be connected with a single cable instead of a bundle of cables 3" thick. Automotive manufacturers found that if a wiring harness is faulty, it is sometimes cheaper to scrap the entire car than to troubleshoot the wiring harness. In a network, we can do wiring in software, and the added hardware cost is more than paid for by labor savings. The same applies to automated equipment in a factory.[16]

4.7.3 Ethernet: Worldwide standard for business and PC Networking

Ethernet is just a physical layer standard, in much the same way an RS232, or a telephone line. The next frontier for the established fieldbus organizations is to produce Ethernet TCP/IP application layers of their protocols. Presently, there are four major contenders: Modbus/TCP (Modbus protocol on TCP/IP), EtherNet/IP (the ControlNet /DeviceNet objects on TCP/IP), Foundation Fieldbus High Speed Ethernet, and Profinet (Profibus on Ethernet). One could propose an infinite number of potential application layer protocols, and in fact right now there are, in addition to the above protocols, a myriad of other, proprietary standards from various vendors.[16]

4.8 Operation of PLC

A PLC is continuously running through its program and updating it as a result of the input signals. Each such loop is called cycle. During each cycle, the processor scans all the inputs and copies their status into the input output address in RAM. As the program is executed the stored input data is read as required from RAM and the logic operation carried out.

The resulting output signals are stored in the reserved input/output section of RAM. At

the end of each program cycle all the outputs are transferred from RAM to the output channel. The output remain their status until the next updating. The process is known as cycle. A scan cycle is the time required for a PLC to scan all the inputs and generate approximate control responses of its outputs. Scan cycle varies with program content and length. A scan can take from 1 to 40 ms.

The sequence can be summarized as-

1. Scan all the inputs and copy into RAM
2. Fetch and Decode and execute all the instructions in sequence, Copy output instructions into RAM.
3. Update all outputs
4. repeat the sequence

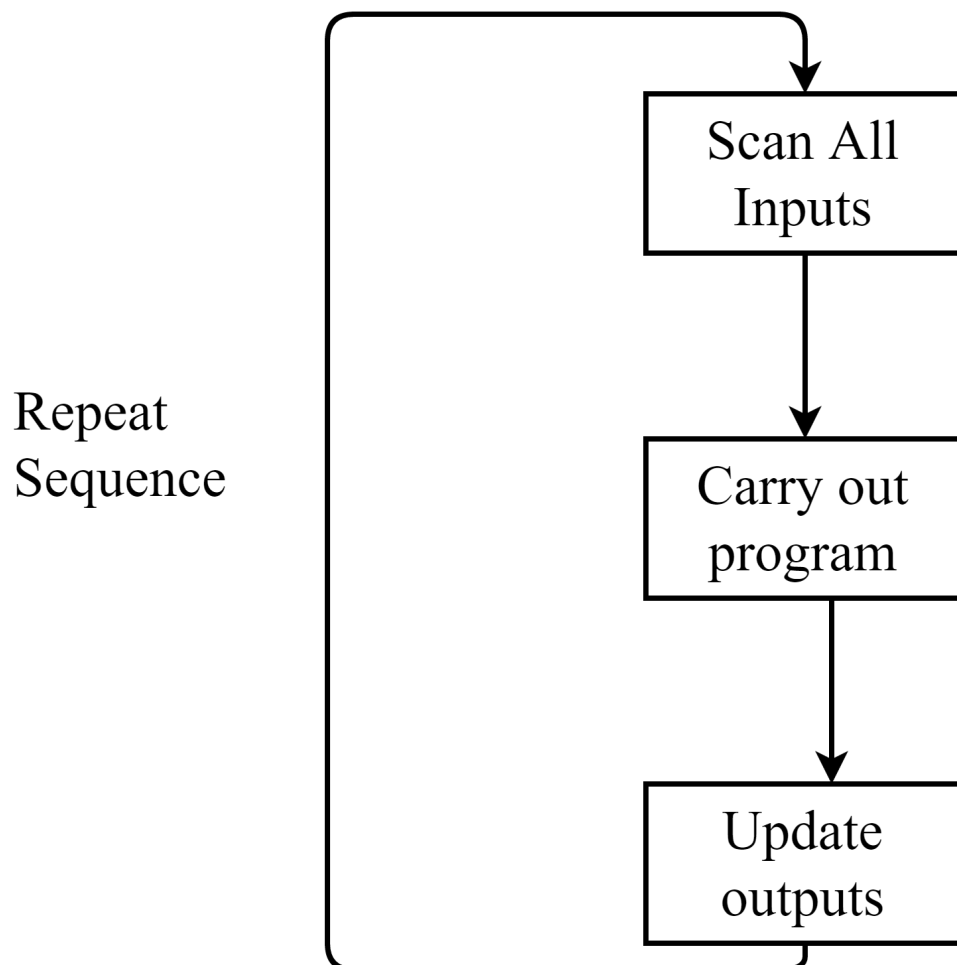


FIGURE 4.9: Operation Cycle of PLC

4.9 PLC Selection

To select the best PLC for any system it is needed to specify the specification of the PLC.

4.9.1 General Specification of PLC

Input and output Specifications

1. No of digital inputs per module and total numbers
2. No of digital output per module and total numbers
3. No of analog inputs per module and total numbers and including input type
4. No of analog outputs per module and total numbers and including output type
5. No of internal outputs or relay.
6. No of special internal outputs.
7. No of internal timers and time set values.
8. No of internal counters and count set values

Other Specification

1. Processing Speed
2. Program capacity
3. Programming system
4. Peripheral Devices
5. Self diagnostics
6. Run/program Mode

Chapter 5

Ladder Logic Diagram And Programming

5.1 Introduction

5.2 Programming Languages of PLC

IEC 1131-3 is the international standard for programmable controller programming languages. The following is a list of programming languages specified by this standard:

1. Ladder diagram (LD)
2. Sequential Function Charts (SFC)
3. Function Block Diagram (FBD)
4. Structured Text (ST)
5. Instruction List (IL)

One of the primary benefits of the standard is that it allows multiple languages to be used within the same programmable controller. This allows the program developer to select the language best suited to each particular task.

5.2.1 Ladder Logic

Ladder logic is the main programming method used for PLC's. Ladder logic has been developed to mimic relay logic. The decision to use the relay logic diagrams was a strategic one. By selecting ladder logic as the main programming method, the amount of retraining needed for engineers and trades people was greatly reduced.

The first PLC was programmed with a technique that was based on relay logic wiring schematics. This eliminated the need to teach the electricians, technicians and engineers how to program - so this programming method has stuck and it is the most common technique for programming in today's PLC.

5.2.2 Mnemonic Instruction

There are other methods to program PLCs. One of the earliest techniques involved mnemonic instructions. These instructions can be derived directly from the ladder logic diagrams and entered into the PLC through a simple programming terminal.

5.2.3 Sequential Function Charts (SFC)

SFC have been developed to accommodate the programming of more advanced systems. These are similar to flowcharts, but much more powerful. This method is much different from flowcharts because it does not have to follow a single path through the flowchart.

5.2.4 Structured Text (ST)

Programming has been developed as a more modern programming language. It is quite similar to languages such as BASIC and Pascal. Structured Text (ST) is a high level textual language that is a Pascal like language. It is very flexible and intuitive for writing control algorithms.

5.2.5 Function Block Diagram (FBD)

FBD is another graphical programming language. The main concept is the data flow that start from inputs and passes in block(s) and generate the output.

5.3 Programming With Ladder Diagram

Ladder Logic programming language, invented in the U.S. decades ago, is probably the most widely used. Invented to replace hardwired relay control systems, Ladder Diagram programming is a mainstay in the U.S. today, used in probably 95 percent of all applications. Visually, this language resembles a series of control circuits, with a series of inputs needing to be made or true in order to activate one or more outputs.

Ladder Diagram language has experienced such widespread adoption that almost every programmer in any country or industry can read and write this language. Because it resembles the familiar electric circuit format, even a non-programmer with an electrical background can follow the program for purposes of troubleshooting a problem. It's also easy to start writing a program in Ladder Diagram. With just a basic outline of input and output signals, one can sit down and start churning out code. Most of the other IEC languages require more preparation, such as flow charting all the potential process flows. Finally, most implementations of Ladder Diagram allow a program to be organized into folders or subprograms that are downloaded to the PLC, allowing for easy segmentation. Ladder Diagram programming is ideal for a simple material handling application, for example, where a sensor detects the presence of a box, other sensors check for obstructions, and then an output fires an actuator to push the box to another conveyor.

Digital inputs are checking for various conditions, and a basic program is analyzing the inputs and firing digital outputs in response. There may be timers in the program, or some basic comparisons or math, but there are no complex functions involved. As the complexity of PLC functionality has grown, however, Ladder Diagram language has been challenged to meet these advances and still maintain the paradigm of easy visualization and understanding. Functions such as PID, trigonometry and data analysis are commonly required in many control applications, but difficult to implement. Another challenge is that as program size grows, the ladder can become very difficult to read and interpret, unless it's extensively documented. Finally, implementing full processes in Ladder Diagram can be a daunting picture a ladder rung with an output used in several phases of a process with many input conditions attempting to control exactly when that output needs to turn on.

5.4 Basic Instructions

Some of the basic instructions are examined below in greater detail to see more about what each one does.

5.4.1 Load

The load instruction is basically a normally open contact. It is sometimes also called examine if on (XIO). The symbol of load instruction is shown in figure 5.1 This is used

NO Contact



FIGURE 5.1: A Load Symbol

when an input signal is needed to be present for the symbol to turn on. When the physical input is on the instruction is true. If the physical input is on the instruction symbol is on. As on state it is refer to as logical 1 state.

5.4.2 Load Bar

The load instruction is basically a normally close contact. It is sometimes also called examine if close (XIC). The symbol of load instruction is shown in figure 5.2 This is used

NC Contact

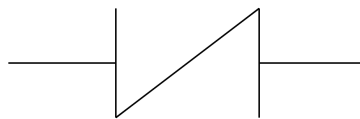


FIGURE 5.2: A Load bar Symbol

when an input signal does not need to be present for the symbol to turn on. When the physical input is on the instruction is false. If the physical input is on the instruction symbol is off. As on state it is refer to as logical 0 state.

5.4.3 Out

The out instruction is sometimes also called an Output energized instruction. The output instruction is likely a relay coil that is normally off. The symbol of out instruction is shown in figure 5.3 When there is a path of true instruction proceeding this on the ladder rung, this is also be true. When the instruction is true it is physically on.

NO Coil

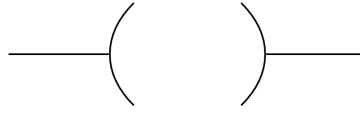


FIGURE 5.3: A out Symbol

5.4.4 Out bar

The out bar instruction is sometimes also called an Output Not energized instruction. The output bar instruction is likely a relay coil that is normally on. The symbol of out instruction is shown in figure 5.9 When there is a path of true instruction proceeding this

NC Coil

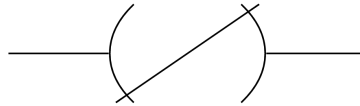


FIGURE 5.4: A out bar Symbol

on the ladder rung, this is also be false. When the instruction is true it is physically off.

5.5 Ladder Logic Inputs

PLC input symbols are not actually physical contacts, They are just instructions existing in a program. A physical input is hardware to the PLC at a specific address. A ladder logic program reference that specific address in a rung. Different PLC software references input address in a various ways. Table 5.1 shows the input address for different PLC module available in market.

5.6 Ladder Logic Outputs

The PLC output symbol is a coil. However, the actual physical output that a PLC program is energized could be a number of things. An output could be anything from an indicator light to a relay to a motor. A PLC does not care what the output is. When the Proper inputs are present and the ladder logic indicates an output should be energized , a PLC sends current to the output.

TABLE 5.1: Address table for Different PLC

PLC	input address	output address	Timer address	Counter address
Simense	I127.0 -I127.7	Q127.0 -Q127.7	T1-T32	C1-C32
omeron	0.00-0.11	10.00-10.7	TIM1#024 -TIM7#027	CNT1#024 -CNT64#028
Delta	X0-X177 Octal System i.e. (X0 To X7,X20 to X27)	Y0-Y177 Octal System	0 to T127	General C0 to C111 , C112 to C127
Allen Bradley	101-109	111-121	Txx xx=0-55	Cxx xx=0-55

Different PLC software references output address in a various ways. Table 5.1 shows the output address for different PLC module available in market.

5.7 Timers

Timers are very important in ladder logic programming. Timers gives the precision in time. Timer on delay starts timing when instruction is true. Timers are used to track time when instruction are on or off. They could also keep track on a retentive base.

There are three types of Timers in PLC. They are:

1. TON - Timer On Delay
2. TOF - Timer Off Dealy
3. RTO - Retentive Timer

5.7.1 TON - Timer On Delay

Count time base intervals when the instruction is true. The Timer On Delay instruction begins to count time base intervals when rung conditions become true. As long as rung conditions remain true, the timer adjust its accumulated value (ACC) each evaluation until it reaches the preset value (PRE). The accumulated value is reset when rung conditions go false, regardless of whether the timer has timed out. Each Timer on Delay is made of a 3-word element.

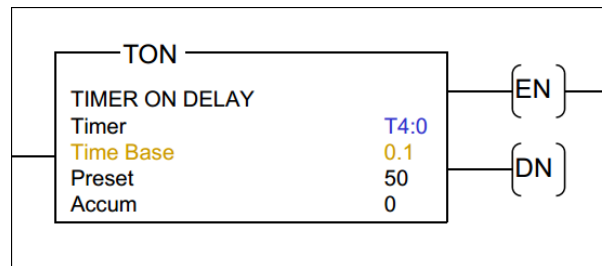


FIGURE 5.5: On delay Timer

1. Word 1 is the control word
 - (a) Bit 0-12: Internal Use
 - (b) Bit 13: Done (DN) this bit is on when the Accumulation value $\geq$ Preset Value
 - (c) Bit 14: Timer Timing (TT) this bit is on when the timer is timing
 - (d) Bit 15: Enabled (EN), this bit is on when the timer is energized.
2. Word 2 stores the preset value. (PRE)
 - (a) The programmer specifies this value. When the accumulated time reaches the preset value the controller sets the done bit. When the accumulated value becomes equal to or greater than the preset value, the done bit is set. Usually preset value is from 0 - 32,767
 - (b) If a timer-preset value is negative an error will occur.
3. Word 3 stores the accumulated value. (ACC)
 - (a) This is the time elapsed since the timer was last reset. When enabled the timer updates this continually.
 - (b) Time Base: is the timing update interval, this can vary from 0 - 1 second.

5.7.2 TOF - Timer Off Delay

Counts time base intervals when the instruction is false. The Timer Off Delay instruction begins to count time base intervals when the rung makes a true to false transition. As long as rung conditions remain false, the timer increments its accumulated value (ACC) each scans until it reaches the preset value (PRE). The accumulated value is reset when rung conditions go true regardless of whether the timer has timed out. Each timer address is made of a 3-word element.

1. Word 1 is the control word

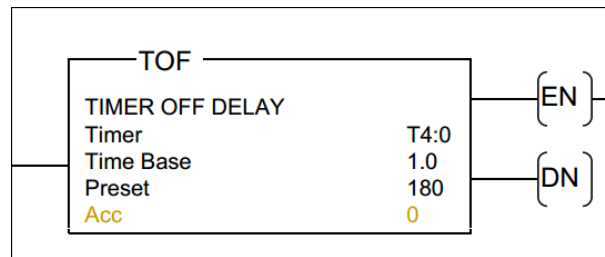


FIGURE 5.6: Off delay Timer

- (a) Bit 0-12: Internal Use
- (b) Bit 13: DN- Done
- (c) Bit 14: TT - Timer Timing
- (d) Bit 15: EN - Timer is enabled
- (e) Word 2 stores the preset value. (PRE)

- (a) Specifies the value, which the timer must reach before the controller sets the done bit. When the accumulated value becomes equal to or greater than the preset value, the done bit is set.
- (b) Preset value is from 0 - 32,767
- (c) If a timer-preset value is negative an error will occur.

2. Word 3 stores the accumulated value. (ACC)

- (a) This is the time elapsed since the timer was last reset. When enabled the timer updates this continually.
- (b) Time Base: is the timing update interval, this can vary from 0 - 1 second.

5.7.3 RTO - Retentive Timer

Counts time base intervals when the instruction is true and retains the accumulated value when the instruction goes false or when power cycle occurs. The Retentive Timer instruction is a retentive instruction that begins to count time base intervals when rung conditions become true. The Retentive Timer instruction retains its accumulated value when any of the following occurs: Rung conditions become false. Changing Processor mode from REM run /Test / program mode. The processor loses power while battery back up is still maintained and a fault occurs. Each Retentive Timer is made of a 3-word element.

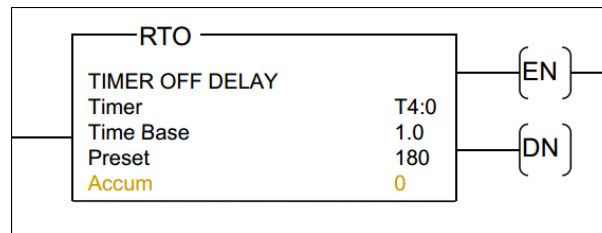


FIGURE 5.7: Retentive Timer

1. Word 1 is the control word
 - (a) Bit 0-12: Internal Use
 - (b) Bit 13: DN- Done
 - (c) Bit 14: TT - Timer Timing
 - (d) Bit 15: EN - Timer is enabled
2. Word 2 stores the preset value. (PRE)
 - (a) Specifies the value, which the timer must reach before the controller sets the done bit. When the accumulated value becomes equal to or greater than the preset value, the done bit is set.
 - (b) Preset value is from 0 - 32,767
 - (c) If a timer-preset value is negative an error will occur.
3. Word 3 stores the accumulated value. (ACC)
 - (a) This is the time elapsed since the timer was last reset. When enabled the timer updates this continually.
 - (b) Time Base: is the timing update interval, this can vary from 0 - 1 second.

5.8 Counters

Counters are very essential in ladder logic programming. Counters are used to index, increment or decrement values. There are three types counter instructions in PLC.[17]

1. CTU - Count Up
2. CTD - Count Down
3. RES - Reset

5.8.1 CTU-Count UP

Increments the accumulated value at each false to true transition and retains the accumulated value when the instruction goes false or when power cycle occurs. The CTU is an instruction that counts false to true transition. When this transition happens the accumulated value is incremented by one count. A CTU accumulation is reset by the RES instruction. If the accumulation value is over the maximum range then the overflow (OV) bit will be true. Each counter address is made of a 3-word element.

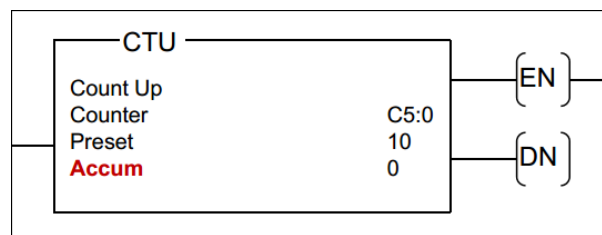


FIGURE 5.8: Counter UP

1. Word 1 is the control word
 - (a) Bit 0-7: Internal Use
 - (b) Bit 10: UA - Update accumulation value.
 - (c) Bit 11: UN - Underflow bit.
 - (d) Bit 12: OV - Overflow bit.
 - (e) Bit 13: DN - Done
 - (f) Bit 14: CD - Count down is enabled.
 - (g) Bit 15: CU - Count up is enabled.
2. Word 2 stores the preset value. (PRE)
 - (a) Specifies the value, which the counter must reach before the controller sets the done bit. When the accumulator value becomes equal to or greater than the preset value, the done status bit is set. You can use this bit to control an output device.
 - (b) Preset value is from -32,768 to 32,767
 - (c) If a timer-preset value is negative an error will occur.
3. Word 3 stores the accumulated value. (ACC)
 - (a) This is the number of times of false to true transitions that have occurred since the counter was last rest.

5.8.2 CTD-Count Down

Decrements the accumulate value at each false to true transition and retains the accumulated value when the instruction goes false or when power cycle occurs. The CTD is an instruction that counts false to true transition. When this transition happen the accumulated value is decrements by one count. A CTD accumulation is reset by the RES instruction. If the accumulation value is below the minimum range then the underflow (UN) bit will be true. Each counter address is made of a 3-word element.

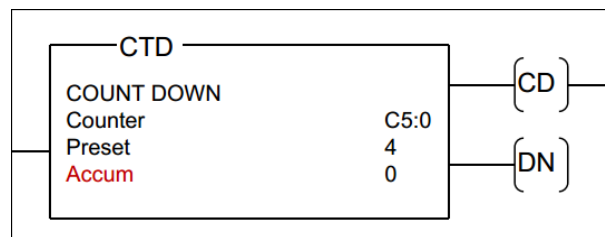


FIGURE 5.9: Counter Down

1. Word 1 is the control word
 - (a) Bit 0-7: Internal Use
 - (b) Bit 10: UA - Update accumulation value.
 - (c) Bit 11: UN - Underflow bit.
 - (d) Bit 12: OV - Overflow bit.
 - (e) Bit 13: DN - Done
 - (f) Bit 14: CD - Count down is enabled.
 - (g) Bit 15: CU - Count up is enabled.
2. Word 2 stores the preset value. (PRE)
 - (a) Specifies the value, which the counter must reach before the controller sets the done bit. When the accumulator value becomes equal to or greater than the preset value, the done status bit is set. You can use this bit to control an output device.
 - (b) Preset value is from -32,768 to 32,767
 - (c) If a timer-preset value is negative an error will occur.
3. Word 3 stores the accumulated value. (ACC)

5.8.3 RES Reset

Resets the accumulated value and status bit of a timer or counter. Use a RES instruction to reset timers or counters. When the RES instruction is enabled, it resets the Timer On Delay, Retentive Timer, and Counter Up, Counter Down instruction having the same address as the RES instruction.

5.9 Data Blocks

Data blocks are an important part of your user program since they contain all its data. Data blocks (DBs) are used to handle data which is why they do not have a code section. Programming data blocks involves the following :

1. **Declaration table:** The declaration table is where user can specify the data structure of the data block.
2. **Block properties:** These include extra information such as time stamp, programming language and path name, which is all entered by the system itself. User can also add information about the name, family, version and author and user can assign system attributes for blocks .

A user program can have the following data blocks:

1. **Shared DBs** can be accessed by all logic blocks in the program. The data remains stored in the data block even when it has been closed.
2. If user require several shared DBs of the same data structure, user can create them with the help of a UDT. These are data blocks with an associated **user-defined data type**.
3. **Instance DBs** are associated with specific function blocks and are structured according to the declaration table of the FB. User can only create an instance DB if the corresponding function block exists. They are data blocks with an associated function block.

5.10 User Function Blocks

The term function block diagram (FBD) is used for PLC programs described in terms of graphical blocks. It is described as being a graphical language for depicting signal and data flows through blocks, these being reusable software elements. A function block is a program instruction unit which, when executed, yields one or more output values. The

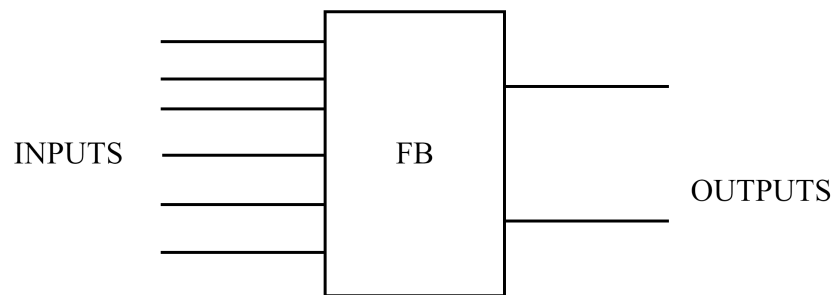


FIGURE 5.10: A Function Block Symbol

IEC 113-3 standard for drawing such blocks is shown in Figure 5.10. A function block is depicted as a rectangular block with inputs entering from the left and outputs emerging from the right. The function block type name is shown in the block, e.g., AND, with the name of the function block in the system shown above it, Timer1. Names of function block inputs are shown within the block at the appropriate input and output points. Cross diagram connectors are used to indicate where graphical lines would be difficult to draw without cluttering up or complicating a diagram and show where an output at one point is used as an input at another.

Function blocks can have standard functions, such as those of the logic gates or counters or times, or have functions defined by the user, e.g., a block to obtain an average value of inputs.[17]

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mass seeking system

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Chapter 10

Conclusion

10.1 Conclusion

10.2 Future Work

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